

What the slack data now says

Five analyses, July 2026 — rundown, the ATP effect, and one recovery rate

1

SlackDataAnalysis

nine recordings, one code path

2

RundownCorrection

what degrades, and how to correct it

3

ATPEffectReconciliation

force $\times 1.36$, ktr $\times 0.55$

4

RestretchVsKtrRecovery

one rate, three model defects

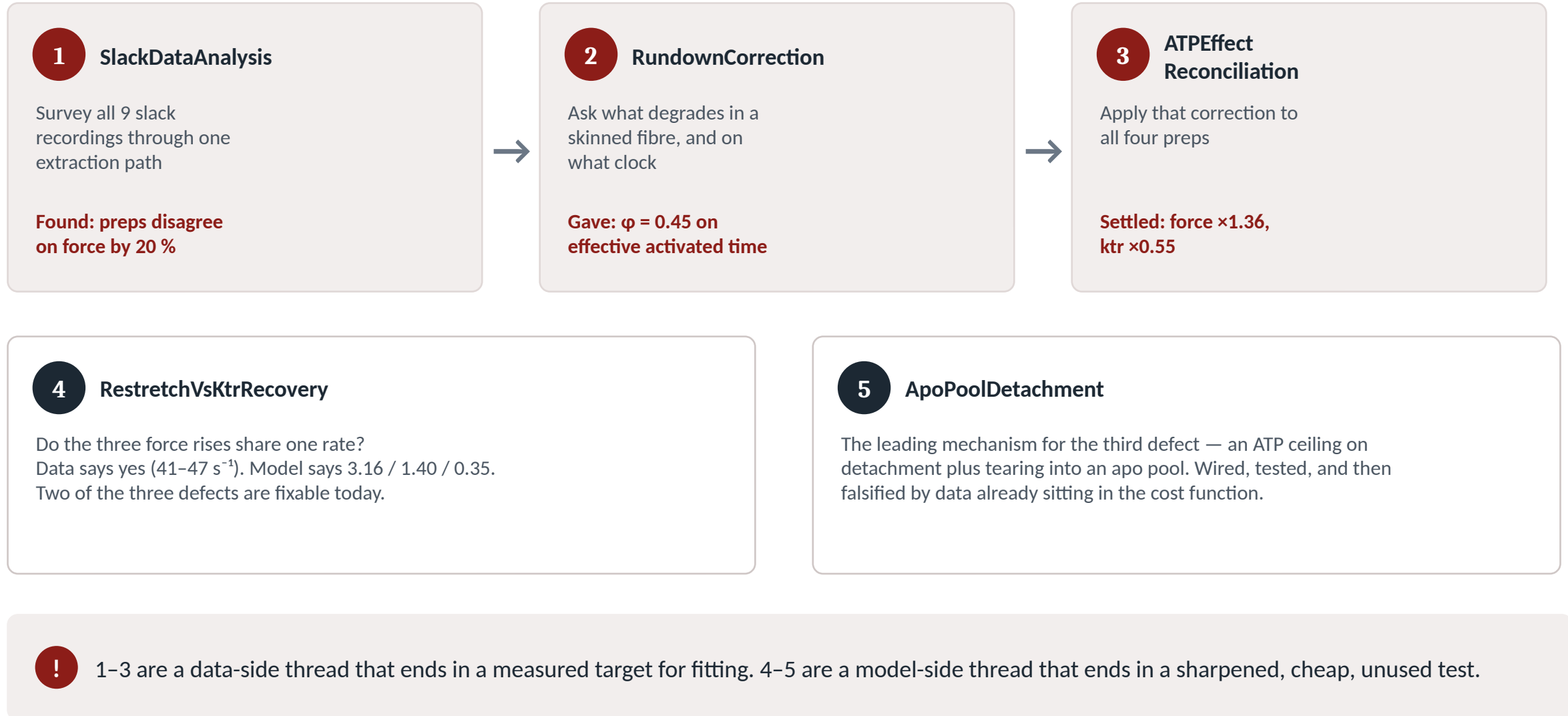
5

ApoPoolDetachment

a candidate mechanism, falsified

All five live in `Analyses/<name>/conclusions.md` — every figure here is that folder's own results/

The five are a chain, not five separate results



The numbers that came out settled

×1.36

isometric force at 2 mM vs 8 mM
ATP

CV 11 %, 3 preps, rundown-corrected

×0.55

k_{tr} , at 2 mM vs 8 mM ATP

CV 8 %, 4 preps, uncorrected

41–47 s⁻¹

one force-redevelopment rate

shared by all three rises, 3 days

$\varphi = 0.45$

permanent fraction of rundown

two independent calibrations agree

and the ones that came out negative — equally useful

- Baker amplitude features are not poolable with the 2026 preps — different protocol, 2–6× shorter recovery, truncated asymptotes.
- Correcting k_{tr} for rundown makes cross-prep consistency worse (CV 8 % → 13–16 %). Leave it raw.
- Rundown is not lost myosin heads — a pure force scale cannot bend the length–tension curve, and the curve bends.
- A cap on strain-dependent detachment (the apo-pool mechanism) is falsified by the k_{tr} overstretch peak.
- $ovrsht_*$ and $ktr2_*$ are retired from the cost function — ratio CV 115 % and 41 %.



SLACKDATAANALYSIS

1

Nine recordings, one extraction path — what can actually be compared?

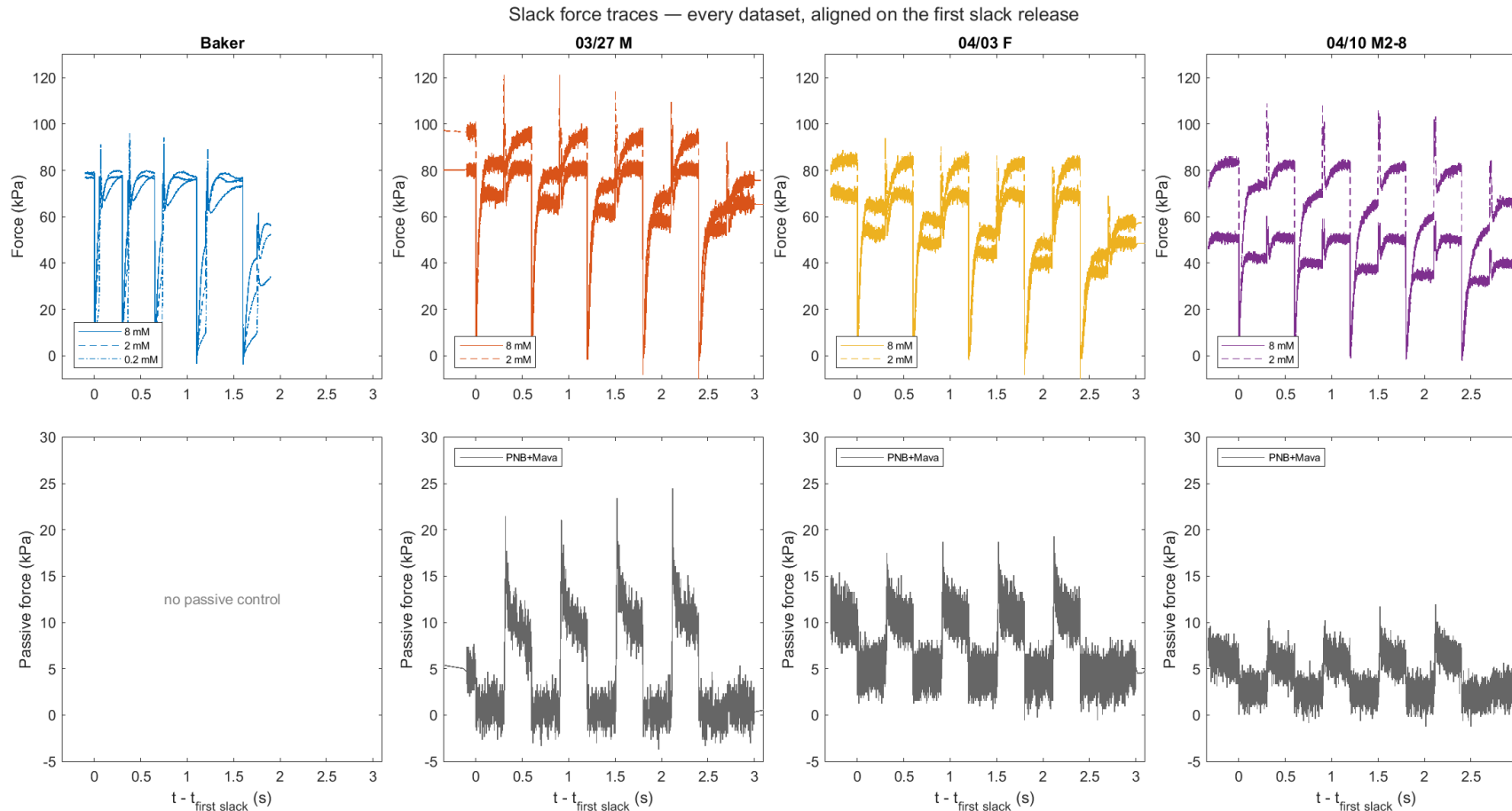
WHY WE RAN IT

- Stored features_data had been written at different times by different script revisions — some smoothed, some not.
- A cross-dataset comparison only means something if every number comes from one code path.
- Fits were being anchored to a single prep's absolute force in kPa, baking that prep's scale into the kinetics.
- The Baker legacy set was being pooled with the 2026 preps without anyone checking the protocols matched.

WHAT IT HAD TO ANSWER

- Which features reproduce across preparations, and which do not?
- Can Baker be pooled with the 2026 protocol days?
- Should we be fitting absolute kPa, or shape?
- What is drift within a single recording doing to the extracted features?

Every slack recording, aligned on the first release



READ IT LEFT TO RIGHT

- Baker's slack holds are visibly shorter — the recovery is cut off before it plateaus.
- Absolute force differs by nearly 2× between preps at the same [ATP].
- Segment 5 is protocol-different by design in every dataset (restretch to 1.00 ML, longer hold).
- Passive level varies 2–3× between days — 03/27 is the stiffest, and it is TNF's default reference.

RunSlackDataAnalysis.m → results/overview_traces.png. Top row: active force. Bottom row: the PNB+Mava passive control for that day.

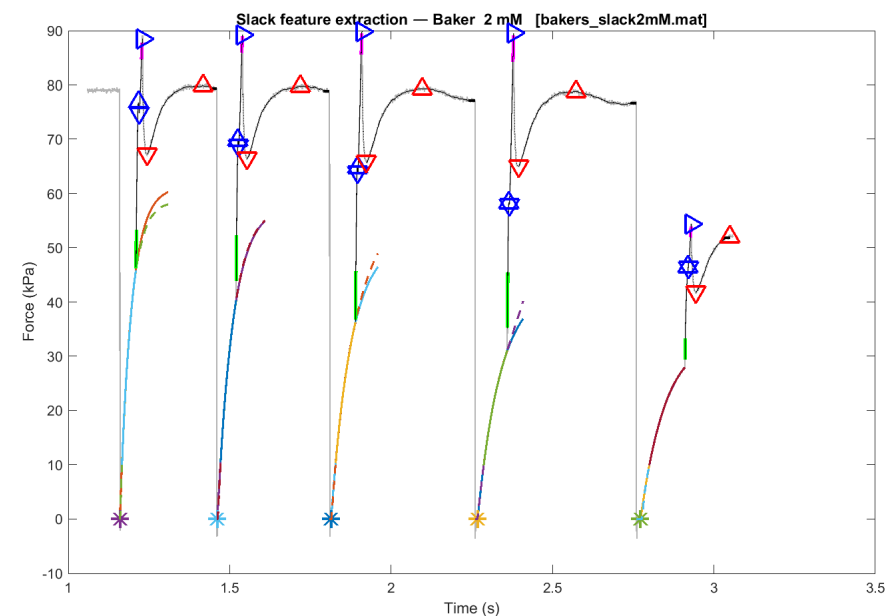
! The three 2026 days share one protocol. Baker does not — and that turns out to matter more than it looks.

Baker is a different experiment — and its recovery is truncated

	Baker	03/27, 04/03, 04/10
inter-slack interval	0.30 / 0.35 / 0.45 / 0.50 s	0.60 s uniform
slack hold = the k_t , fit window	0.05 → 0.15 s	0.298 s
release velocity	-89 ... -174 ML/s	-36 ... -71 ML/s
restretch velocity	4, 5, 6, 7, 3 ML/s	same ✓
$k_t \times \text{hold}$ (>3 ⇒ complete)	1.2 - 3.7	4.0 - 16.7
A_m / A (fraction of asymptote reached)	0.67 - 0.96	0.94 - 1.05

The artifact this manufactures

- k_t is $\sim 0.55\times$ at low ATP, so a fixed hold truncates the low-ATP recovery harder.
- Baker 2 mM / 8 mM: $A_m = 0.713$, $A = 0.856$ — an apparent force LOSS at low ATP...
- ...but steady = 1.006 and peak2 = 1.083, read long after the restretch, show no loss at all.
- The effect appears in exactly the truncation-sensitive features and in no others.

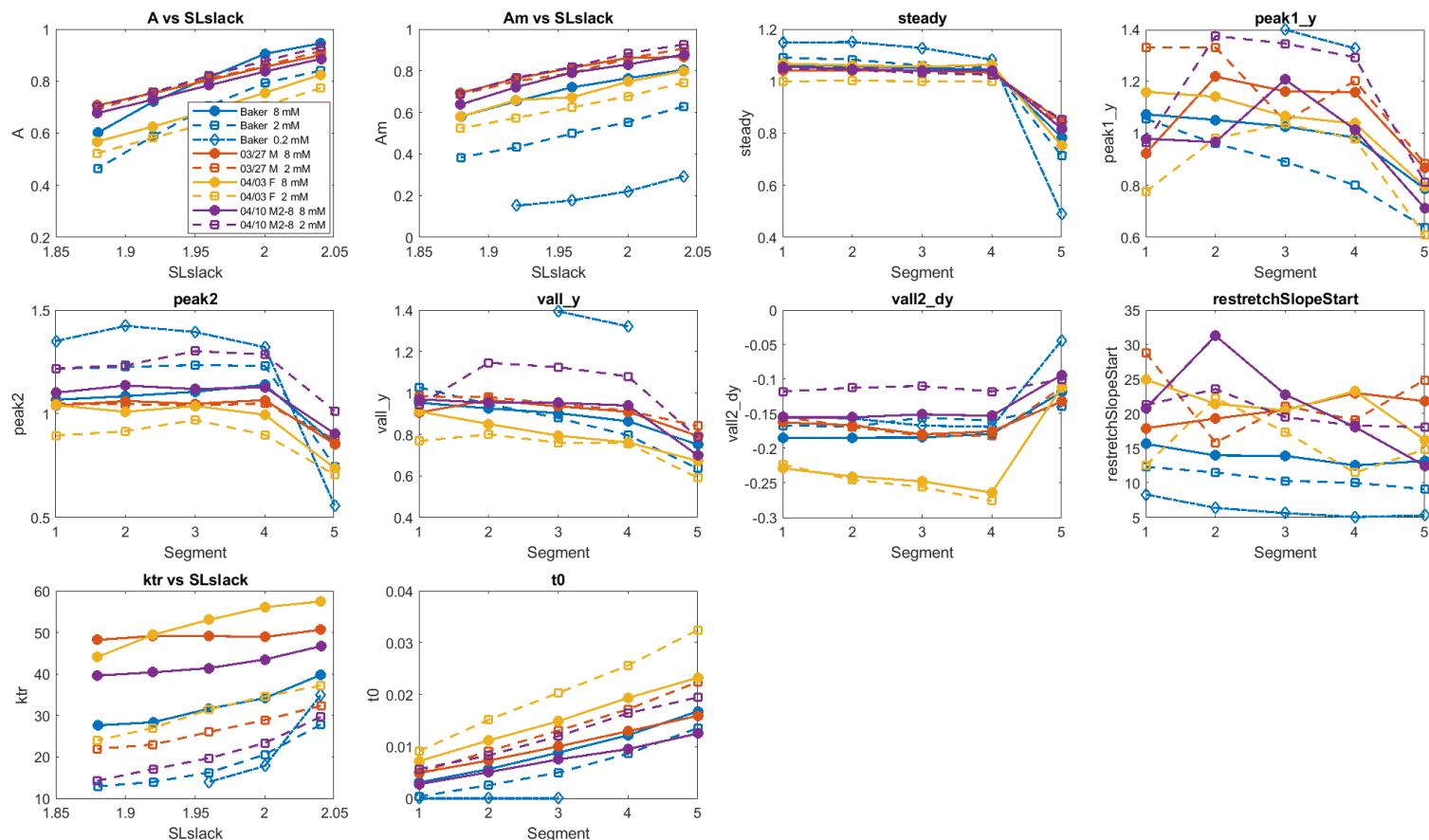


Baker 2 mM QC fit — the exponential fits are still climbing when the window ends.



Do not pool Baker amplitudes with the 2026 preps. Its k_t is fine — a rate is identifiable from 1.2τ , an asymptote is not.

Absolute force is not transferable. Shape is.



Force features divided by each prep's own mean steady. Colour = day; solid = 8 mM, dashed = 2 mM, dash-dot = 0.2 mM.

feature	raw CV	normalised
A / Am	23 / 22 %	7 / 7 %
peak1_y	22 %	4 %
peak2	19 %	5 %
vall_y	21 %	6 %
vall2_dy	28 %	19 %
restretchSlopeStart	23 %	19 %

- Absolute 8 mM A spans 37.5 – 62.2 kPa across preps — cross-section, damage and rundown, not model error.
- Fit normalised shape plus ONE free per-prep force scale.
- Fitting absolute force to a single prep is what ThursdayNightFever does: it matches 03/27's A to 0.1 % and misses 04/10 by 1.6–2×.



Between-prep CV falls from ~20 % to 4–7 % the moment each prep's own strength is divided out.

Which features to trust — and what changes in the cost function

USE

SLslack (CV 0 %) · k_{tr} ratio (8 %) · normalised peak1_y, peak2, vall_y, A, A_m (4–7 %)

DE-WEIGHT

t_0 · restretchSlopeStart · vall2_dy (19–37 %)

DO NOT FIT

ovrsht_dy — ratio CV 115 %, sign flips (+1.35 / +0.34 / +0.37 / -0.01)

$k_{tr2_}^*$ — k_{tr2_k} CV 41 %; $k_{tr2_omega} = 158$ for Baker 2 mM. The 2-component fit is not identifiable on this data.

Six changes to how we fit

- 1. Fit ratios, not absolute levels
- 2. Fit normalised shape + one per-prep scale
- 3. Exclude Baker amplitudes; keep its k_{tr}
- 4. Retire $ovrsht_^*$ and $k_{tr2_}^*$
- 5. Detrend every recording by its own slope
- 6. Re-measure or de-weight 04/03; get a per-day passive



Detrending matters: each recording decays during its own activation, and the 2 mM runs decay 1.5–2× faster, so drift does not cancel in the ATP ratio — it is 8–16 % of the apparent A(1→5) decline.

2

RUNDOWN CORRECTION

What actually degrades in the preparation — and on what clock

WHY WE RAN IT

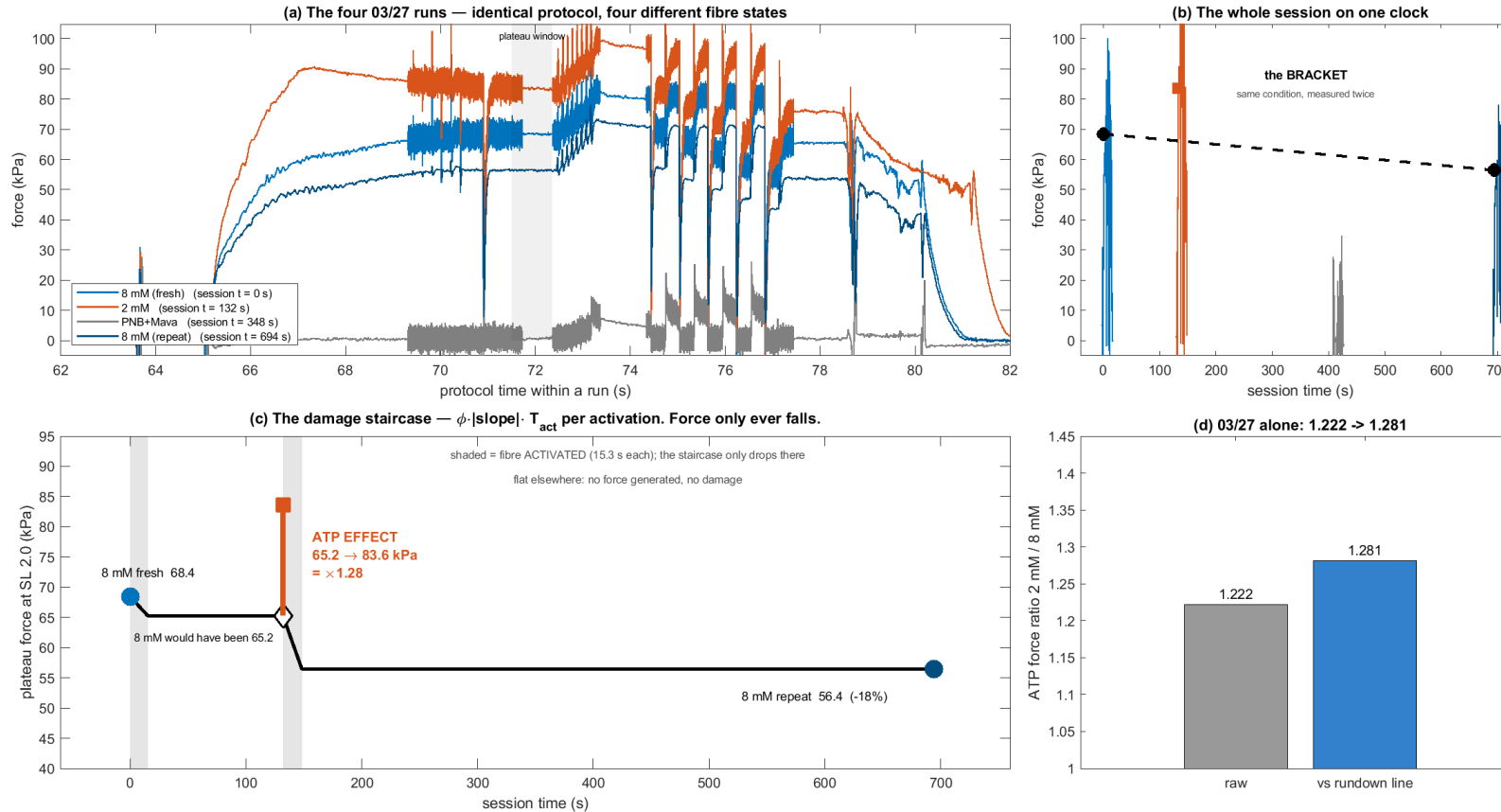
- The survey left the preps disagreeing on the ATP force effect by CV 20 % while agreeing on k_{tr} to 8 %.
- That pattern is the signature of rundown, not biology: whichever condition is recorded second is degraded.
- The earlier treatment used wall-clock time and an exponential decay — it under-corrected and labelled 04/10 an irreconcilable outlier.
- 03/27 recorded 8 mM twice, fresh and at the end of the session. That bracket measures rundown with no modelling assumptions.

WHAT IT HAD TO ANSWER

- Is the clock wall-clock time, or accumulated activated time?
- Is the decay exponential or linear?
- How much of each run's within-run decline is permanent?
- Mechanically, what is the lesion — fewer heads, lost length, slower kinetics, or something else?
- Should k_{tr} be corrected too?

The bracket: the same 8 mM protocol, run twice

What "the bracket" is: the same 8 mM protocol run twice, so rundown can be seen without ATP



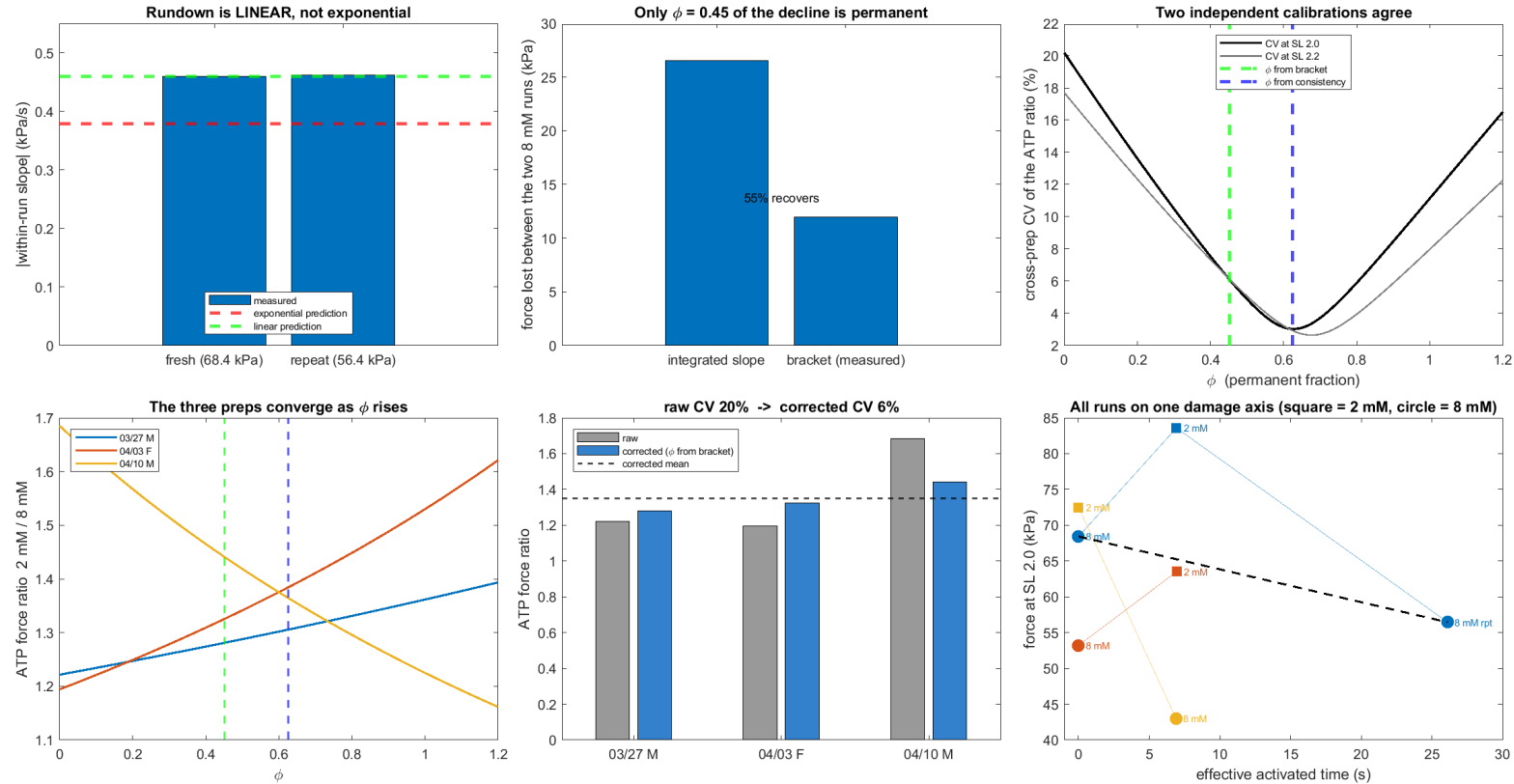
RunBracketExplainer.m $\rightarrow$ results/bracket_explained.png

"Bracketed" means bracketed in TIME, not in force. Force only ever falls: 68.43 $\rightarrow$ 56.45 kPa (-17.5%). The 2 mM run reads higher than both only because low ATP increases force.

! Two measurements of the identical condition, 694 s apart, with a known activation history in between — that is the only assumption-free measurement of rundown in the dataset.

Linear in activated time, and only 45 % of it is permanent

Rundown correction on effective activated time — 03/27 8 mM bracket calibrates it



The slope is force-independent

-0.4592 kPa/s fresh vs -0.4618 kPa/s at -17.5 % force.
Exponential decay predicted -0.379.

$\phi = 0.45$ is permanent

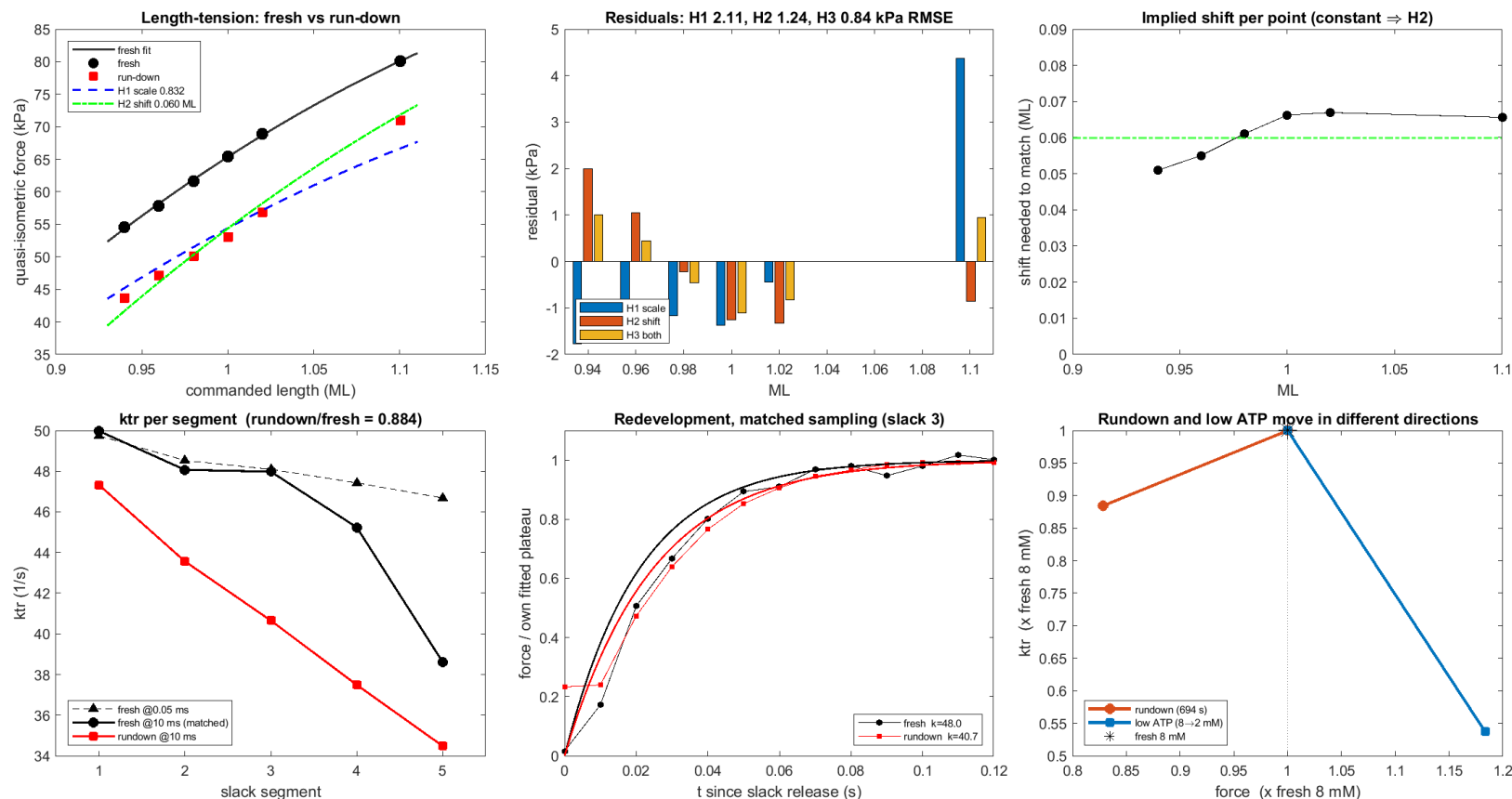
Integrating each run's slope over its 15.3 s activation predicts 26.5 kPa of loss; the bracket measures 11.98. ~55 % recovers between activations.

Two calibrations, one answer

Bracket (mechanical, no ATP assumption) gives 0.452; minimising cross-prep CV of the ATP ratio gives 0.625. Across that whole range the corrected effect is $\approx \times 1.35$.

A scale cannot bend the curve — and the curve bends

Rundown mechanism — 03/27 8 mM fresh ($t=0$) vs repeat ($t=+694$ s)



fit to the L-T curve	RMSE	Δ AIC
pure vertical SCALE (fewer heads)	2.11 kPa	+9.0
pure horizontal SHIFT (lost length)	1.24 kPa	+2.6
BOTH ($s = 0.944$, $d = 0.044$ ML)	0.84 kPa	0

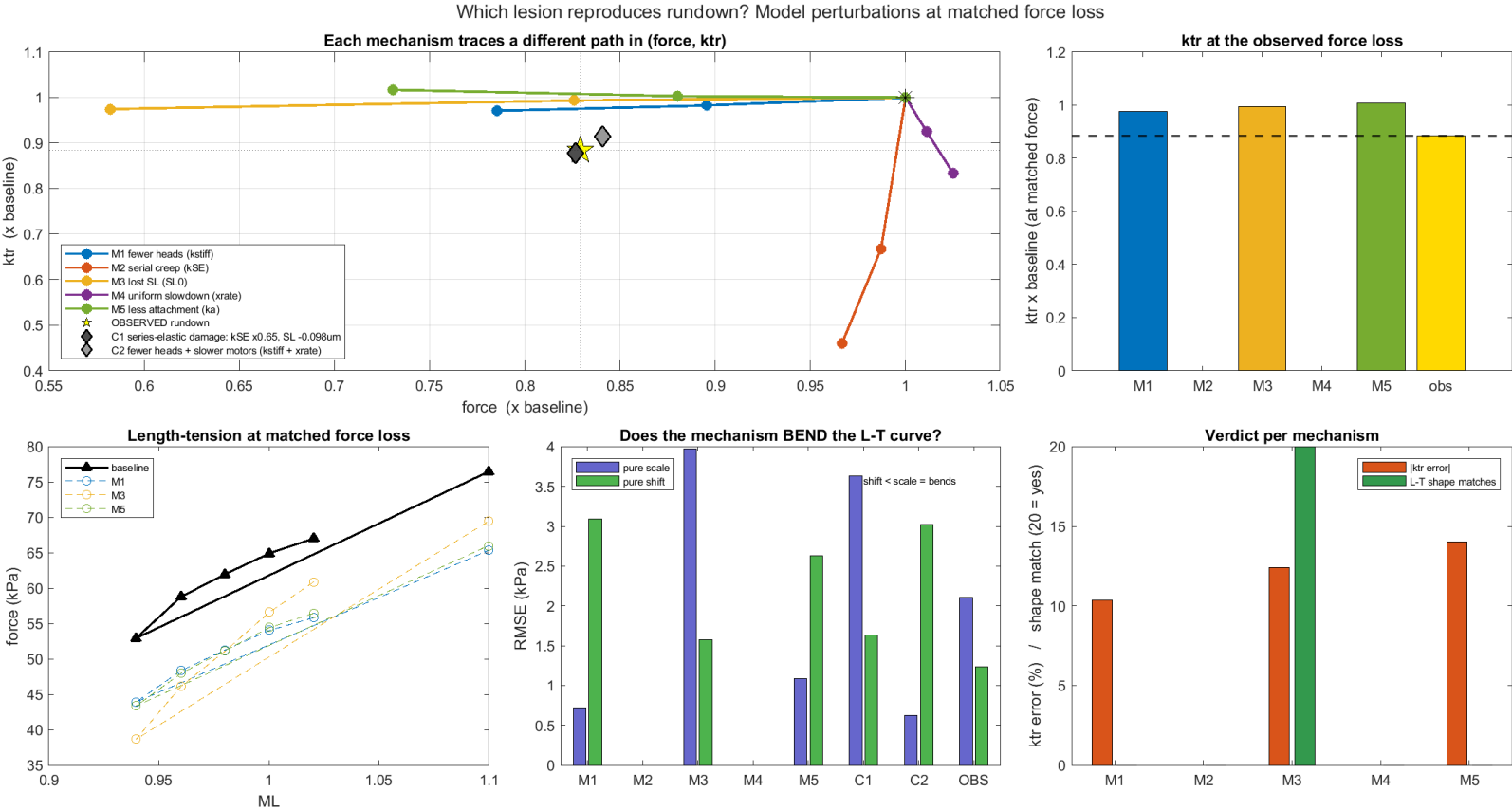
- A pure scale is refuted by the ML 1.10 point alone: it needs +4.4 kPa there while every other point sits ~1.2 kPa low.
- k_{tr} , sampling-matched: $\times 0.884 \pm 0.021$, all five segments down, $t = 5.5$. Sampling alone accounts for only $\times 0.956$.
- Rundown and low ATP move in different directions in the (force, k_{tr}) plane — bottom-right panel.

RunRundownMechanisms.m. Each recording contains a 6-point length-tension curve: the five slack holds (ML 0.94–1.02) plus the pre-slack plateau at ML 1.10.



The discriminator is the SHAPE of the length-tension curve, not the force loss.

Five lesions imposed on the model — none works alone



	force	k _t	L-T
M1 fewer heads	—	0.970	scale ✗
M3 lost SL	—	0.982 ✗	SHIFT ✓
M5 less attachment	—	1.045 ✗	scale ✗
C1 series-elastic damage	0.827	0.877	SHIFT ✓
C2 heads + slower motors	0.841	0.914	scale ✗
OBSERVED	0.829	0.884	SHIFT

M2 (serial creep, kSE) and M4 (uniform slowdown, xrate) barely move force at all — they are pure k_t levers. But M2 and M3 are the SAME physical lesion: a series elastic element that creeps longer and softer lets the sarcomeres sit shorter at a given ML and adds compliance.

RunMechanismSimulation.m — each lesion interpolated to the level that reproduces the observed force loss exactly, so k_t , and the L-T shape become free discriminators.



Verdict: rundown is series-elastic creep — kSE x0.65 and SL -0.098 μ m reproduces force x0.83, k_t x0.88 and the L-T bend. ~35 % softer, ~0.1 μ m longer — not fewer myosin heads.

Do not correct k_{tr} — and what to do instead

Judged on whether it improves cross-prep consistency, the k_{tr} correction makes things worse

k_{tr} ratio 2 mM / 8 mM	03/27	04/03	04/10	Baker	CV
raw	0.537	0.593	0.492	0.566	8 %
rundown-corrected	0.550	0.607	0.465	—	13–16 %

The bracket's k_{tr} loss does not transfer to the short gaps. Over a 132–138 s gap the bias is ~2 %. Leave k_{tr} raw — and never compare k_{tr} across long session gaps.

NEXT MEASUREMENTS

- Record a hi-res *_slack.txt on the end-of-session repeat (mergeLogsAndBursts currently excludes “repeat” filenames — a one-line change).
- Measure stiffness directly: $\text{restretchSlopeStart} / v_{\text{restretch}}$ separates “fewer heads” from “series creep” in one number.
- Repeat the bracket on a second prep to test whether $\varphi \approx 0.45$ is universal.

1 Correct force

$\text{loss} = \varphi \cdot |\text{slope_earlier}| \cdot T_{\text{act_earlier}}$, $\varphi = 0.45$, referenced to the LATER run's own force.

2 Do not correct k_{tr}

It degrades consistency under both the wall-clock and the effective-time model.

3 Prefer a length offset to a force scale

When representing rundown in the model. A consequence: the nominal SLslack axis (2.04 ... 1.88 μm) is systematically too long for later activations.

4 FitRundownCorrection.m's $r(F,SL)$ stays

And §4 explains why it HAS to be SL-dependent. Its hand-tuned $r_0 = 1.214$, $k = -0.6$ reproduces the bracket to within 2 %.

3

ATPEFFECTRECONCILIATION

What 2 mM ATP actually does, across all four preparations

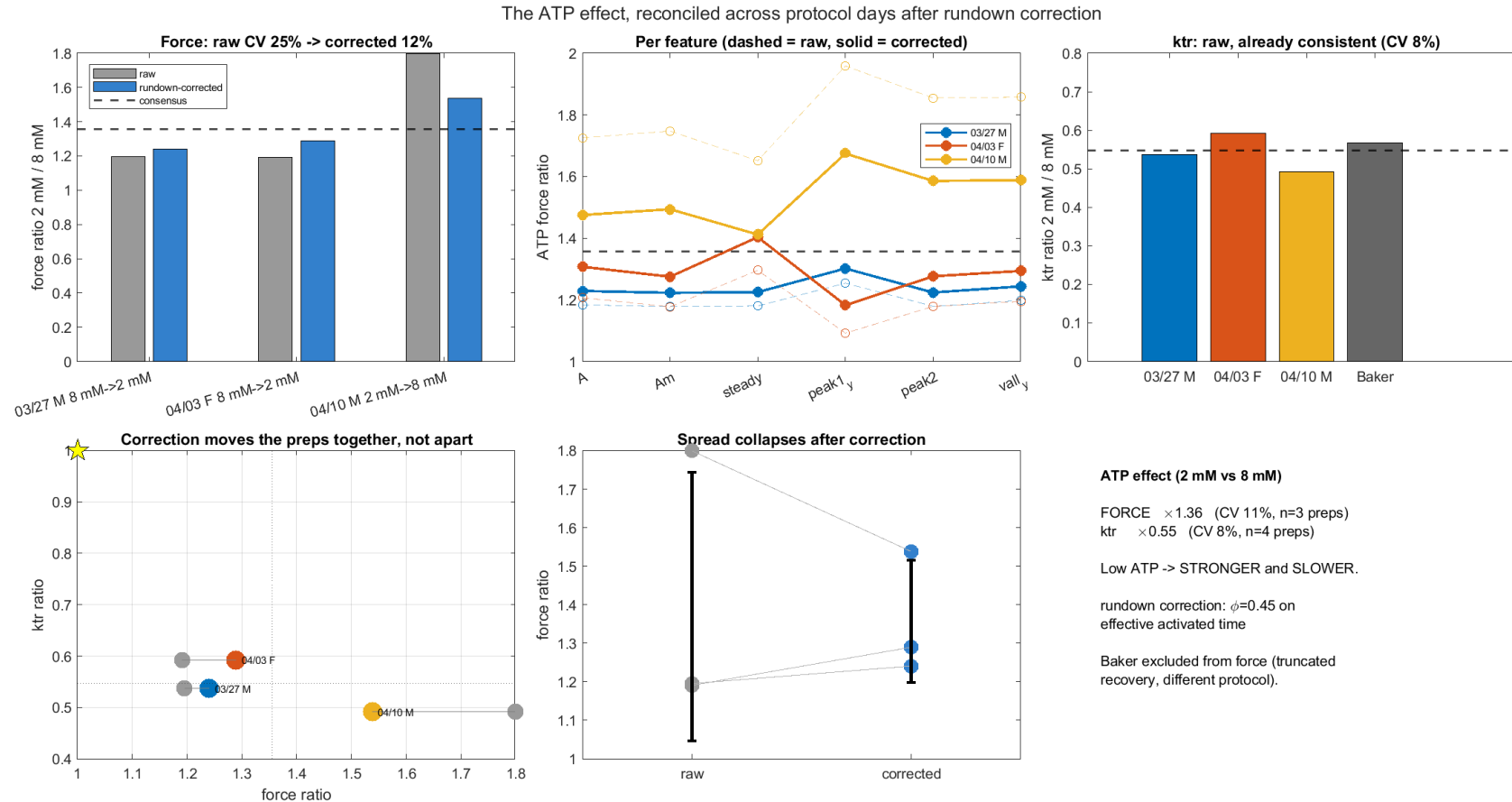
WHY WE RAN IT

- Raw, the four days disagree badly on force ($\times 1.18$ to $\times 1.73$) while agreeing tightly on k_{tr} ($\times 0.49$ to $\times 0.59$).
- Three days ran 8 mM $\rightarrow$ 2 mM. 04/10 ran 2 mM $\rightarrow$ 8 mM — so it carries the opposite rundown bias.
- Earlier numbers in circulation ($\times 1.18$, $\times 1.23$) were either uncorrected or wall-clock-corrected.
- The optimiser needs a target. Without a reconciled number there is nothing to fit the ATP effect to.

WHAT IT HAD TO ANSWER

- Does the rundown correction actually pull the preps together, or apart?
- What is the consensus force ratio, and how confident are we?
- Does k_{tr} need correcting?
- Is 04/10 still an outlier after correction?

Correction moves the preps together, not apart



RunATPReconciliation.m → results/atp_reconciliation.png. Dashed = raw, solid = rundown-corrected.

Correction size per day: 03/27 3.8 % · 04/03 8.3 % · 04/10 17.0 % (reversed order)



Force CV 22 % → 11 %. All six force features move the same way on all three preps. k_{tr} was already consistent at CV 8 % and was deliberately left alone.

Low ATP makes the muscle stronger and slower

feature	raw 03/27 / 04/03 / 04/10	raw CV	corrected	CV
A	1.184 / 1.207 / 1.725	22 %	1.229 / 1.307 / 1.475	9 %
Am	1.178 / 1.177 / 1.747	24 %	1.222 / 1.275 / 1.494	11 %
steady	1.180 / 1.296 / 1.651	18 %	1.225 / 1.403 / 1.412	8 %
peak1_y	1.254 / 1.092 / 1.959	32 %	1.301 / 1.182 / 1.674	19 %
peak2	1.179 / 1.178 / 1.855	28 %	1.223 / 1.276 / 1.586	14 %
vall_y	1.198 / 1.194 / 1.858	27 %	1.243 / 1.294 / 1.588	14 %
ALL FORCE FEATURES		22 %	×1.36	11 %

Caveats — state these when quoting the number

- Baker is excluded from the force number (truncated recovery). Its k_{tr} is included and it agrees.
- 04/10 remains the highest even after correction (×1.54 vs ×1.24 / ×1.29). Its 8 mM run is the weakest 8 mM recording in the whole set — so ×1.36 is, if anything, an OVER-estimate. The two 8→2 preps alone give ×1.27.
- The correction assumes $\phi = 0.45$ transfers between preps. It is calibrated on one bracket.

WHY THIS IS PHYSIOLOGICALLY SENSIBLE

Fewer ATP molecules → slower ADP release and detachment → bridges dwell longer in force-bearing states → more force per bridge-cycle, but a slower cycle.

force × k_{tr} = ×0.74 — the slowed cycle outweighs the extra force. Matches Beard 2022 in direction, with a larger force effect.

FOR FITTING

- Target force ×1.36 and k_{tr} ×0.55 as RATIOS, with a free per-prep force scale — never two absolute levels.
- Use the rundown-corrected force ratio; use the RAW k_{tr} ratio.
- Weight k_{tr} above force: 8 % CV uncorrected vs 11 % after a model-dependent correction.
- For a single-prep fit, use 03/27 — the only day with the bracket, so its correction is measured, not assumed.

! This supersedes the ×1.18 in LowATP_ForceEnhancement (uncorrected 03/27 only) and the ×1.23 briefly quoted from a wall-clock rundown model.

4

RESTRETCHVSKTRRECOVERY

Do the three force rises share one rate? And why does the model say no?

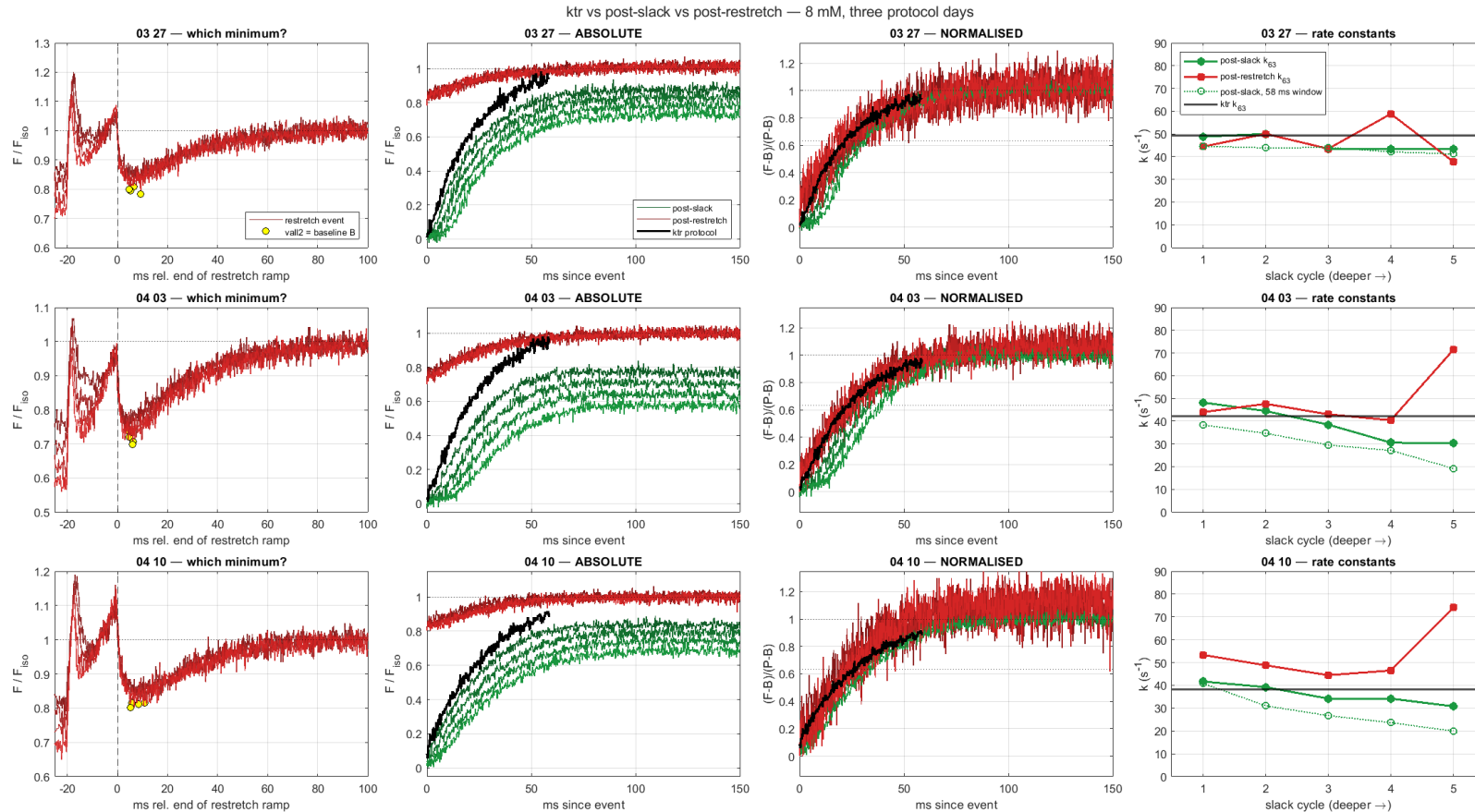
WHY WE RAN IT

- Each slack cycle gives three separate force rises: after the slack release, after the restretch, and in the k_{tr} protocol.
- The model was treating k_{tr} as if it were a different process, and the cost function never scored the comparison.
- An earlier claim that post-restretch is 1.18× slower came from a free-offset fit started at the wrong point.
- Protocol-independence of a redevelopment rate is a sharp, cheap test — if the data has one rate, the model must too.

WHAT IT HAD TO ANSWER

- Do the three windows share one rate constant in the data?
- If so, what does the model give instead?
- Which model components cause the discrepancy?
- Which of them are fixable with flags and parameters already present?

Yes — one rate, $\sim 41\text{--}47\text{ s}^{-1}$, across three protocol days



CompareRecoveryRates.m → results/recovery_rates_traces.png

$k_{63}\text{ (s}^{-1}\text{)}$	03/27	04/03	04/10	mean
post-slack	46.4	40.4	37.3	41.4
post-restretch	49.2	43.7	48.3	47.1
k_{tr} , protocol	49.4	42.1	38.1	43.2

- The between-window spread (41–47) is smaller than the between-day spread (37–49).
- k_{tr} / post-slack = 1.04, and the per-day k_{tr} tracks that day's post-slack value.
- Expected: the k_{tr} manoeuvre ends on a release (1.05 → 1.00 ML), so it starts from zero force at the reference length — exactly like the slack hold.
- Correction to an earlier claim: with the baseline anchored at the true minimum, post-restretch is if anything slightly FASTER, not 1.18× slower.



k_{tr} is not a different process from post-slack redevelopment.

Two estimator traps that produced the wrong answer

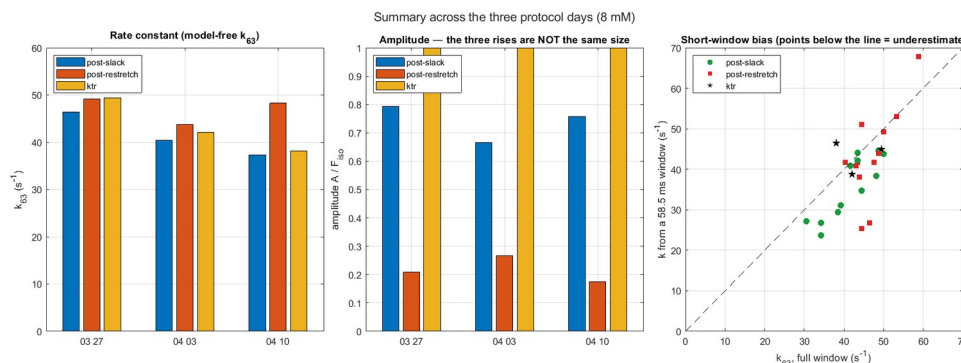
TRAP 1 • which minimum?

peak1 ~1.20 F_{iso} , during the ramp
 → vall_y first dip, still during the ramp
 → peak2 at the ramp end, ~1.00 F_{iso}
 → vall2 ~0.81 F_{iso} , ~7 ms AFTER the ramp ends
 → monotone rise to steady

- The restretch produces TWO valleys, not one.
- Starting the fit at the ramp end means the first ~7 ms are still falling — a free-offset fit absorbs that descent into the baseline and reports a slower rate.
- vall_y (0.96 F_{iso}) is wrong the other way: it is not the start of the recovery, and it understates the amplitude by 4×.
- Use vall2 = min over the 80 ms after the ramp ends.

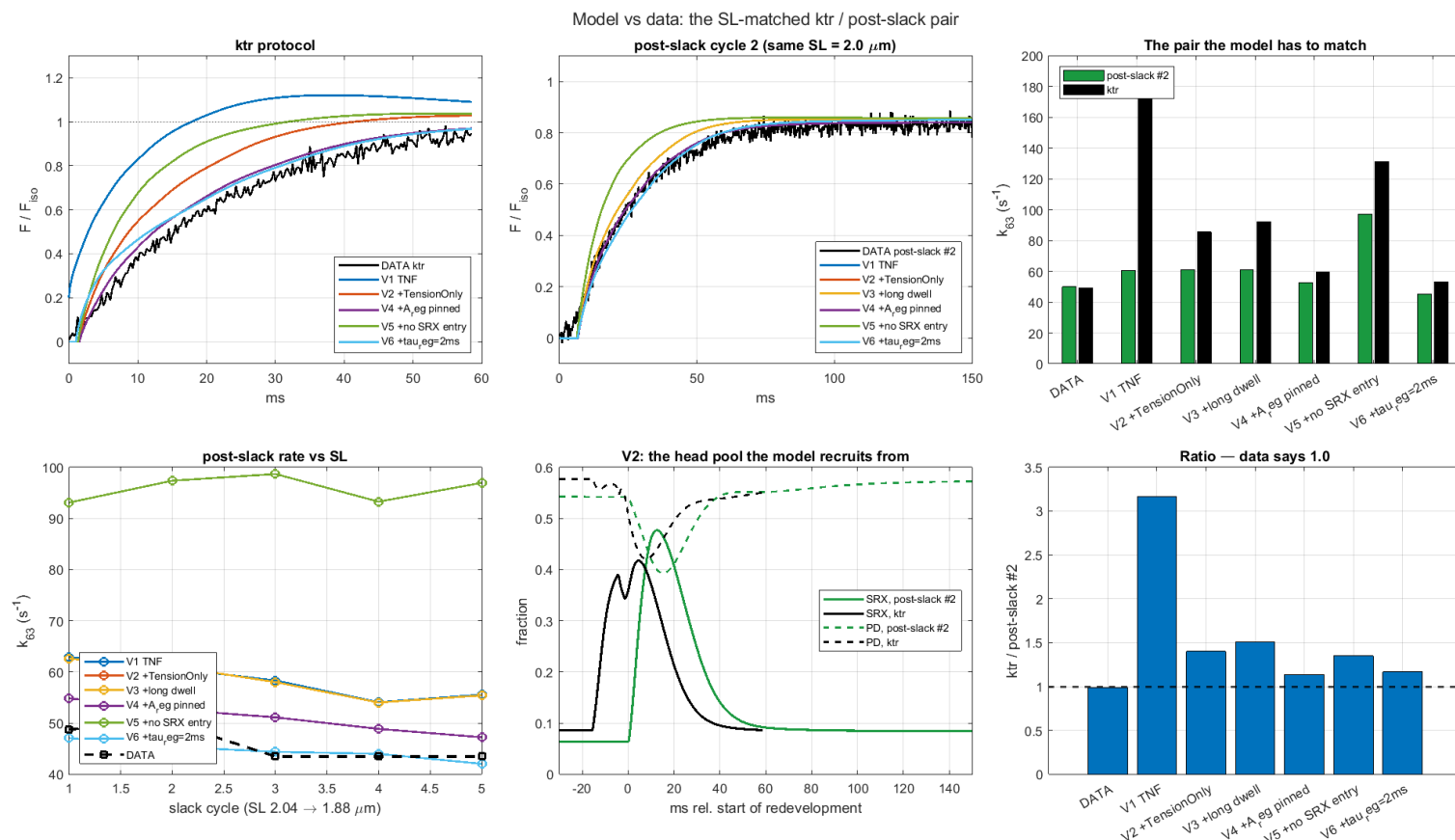
TRAP 2 • the k_t window is truncated

- The k_t window is only 58.5 ms and the trace has reached 0.93 F_{iso} by its end.
- Do NOT let A float in $A(1-\exp(-kt))$ on a k_t recording: refitting post-slack over a matched 58.5 ms span with A free returns only 86 % of the true rate (on 04/03 cycle 5 it collapses to 19 s^{-1} against a true 30).
- Instead fix A at F_{iso} — the muscle returns to the same length, so the plateau is known — or use k_{63} with $P = F_{iso}$.
- extractSlackAttributes' k_t fit anchors $F_0 = 0$ but leaves A free. Safe on the ~275 ms slack windows; not safe on a 58.5 ms k_t window.



Note panel 2: the three rises are NOT the same size — post-restretch is a 0.22 F_{iso} excursion against 0.74 and 1.00. Normalising rescales it 4.6× and its SNR is ~7 against ~30.

The model splits one rate into three



	slack	k_{tr}	ratio	E_{ktr}	E_{slk}
DATA 03/27	50.0	49.4	0.99	—	—
V1 TNF as-is	60.7	192.0	3.16	0.98	183
V2 +TensionOnly	61.1	85.5	1.40	0.23	173
V6 V2 +tau_reg 2 ms	45.4	53.0	1.17	0.05	180

Same length, same starting force (zero), same endpoint — so the two rates must be equal. In the data they are (ratio 0.99). The unmodified model gives 3.16.

THREE DEFECTS ISOLATED

- A titin-dashpot force floor unique to the k_{tr} protocol
- τ_{reg} sitting inside the k_{tr} timescale
- Post-restretch recovery $\sim 2\times$ too fast

ModelVsDataRecovery.m, base params/ThursdayNightFever.m. The test is the SL-matched pair: slack cycle 2 holds at exactly SL 2.0 μm , the length the k_{tr} protocol redevelops at.



Protocol-independence of the redevelopment rate is a sharp, cheap test that the current cost function does not score at all.

Two defects are free to fix. The third is not.

1

A titin-dashpot force floor that only exists in the k_t protocol

UseMaxwellTensionOnly is never set in ThursdayNightFever.m, so it takes the getParams default false. The Maxwell element only ever charges, and the k_t protocol's final 1.05 $\rightarrow$ 1.00 ML release cannot discharge it. At the first sample of the redevelopment window: Force = 12.85, active F_{XB} = -2.47, passive = 16.30 — of which $F_{Maxwell}$ = 16.26. The trace starts at 0.18 F_{iso} where the data starts at 0.000.

FIX: set the flag. k_t , 192 $\rightarrow$ 85.5 s^{-1} , E_{ktr} 0.98 $\rightarrow$ 0.225, and E_{slack} 183.4 $\rightarrow$ 173.4 (also better). FV and stairs unchanged. Strictly an improvement — take it before anything else.

2

τ_{reg} sits inside the k_t timescale

A_{reg} is charged by shortening and relaxes with τ_{reg} = 27.4 ms. The k_t manoeuvre shortens for 6.5 ms; the slack release for 2.2 ms. With a 27 ms memory the model still remembers the difference when redevelopment starts — A_{reg} 0.341 vs 0.284, i.e. 20 % more availability, 20 % more k_a_{eff} , a faster k_t .

FIX: make A_{reg} fast, not slow. At $\tau_{reg} \approx 1$ –2 ms the gate tracks velocity quasi-instantaneously — the FV shoulder survives (it is a steady-state-during-a-ramp property, not a memory) but nothing is remembered once the isometric hold starts.

THE TRADE-OFF, STATED HONESTLY

E_{slack} goes 173.4 $\rightarrow$ 180.1 (τ_{reg} 2 ms) $\rightarrow$ 184.5 (1 ms), and its minimum is at $\tau_{reg} \approx 10$ ms (169.1). τ_{reg} is precisely the parameter that trades slack trace shape against k_t protocol-independence — and because defect 1 kept E_{ktr} pinned at 0.98, the fit had no counterweight and settled at 27 ms. Free τ_{reg} in the next round with the k_t trace properly weighted. Do NOT pin A_{reg} .

DEFECT 3 — STILL OPEN

Post-restretch is $\sim 2\times$ too fast (model 73–100 s^{-1} vs data 44–59) and is slightly WORSENERD by the τ_{reg} fix. $R^2 = k2\cdot p2\cdot ppval(PieceWiseStrainDep2, \cdot)$ reaches 2100–4250 s^{-1} at restretch strains, so every stretched bridge is gone in <0.5 ms and the recovery is pure fast re-attachment.

5

APOPOOLDETACHMENT

The leading mechanism for defect 3 — built, tested, and falsified

WHY WE RAN IT

- Defect 3 needs new mechanism: neither the Maxwell fix nor τ_{reg} touches the post-restretch rate.
- $R2(s)$ reaching 4600 s^{-1} conflates two physically distinct pathways — chemical cycle completion and mechanical rupture.
- The obvious candidate: cap detachment (ATP-limited) and give torn bridges somewhere to go.
- It was worth wiring properly so it could be tested, and so a negative result would be defensible.

WHAT IT HAD TO ANSWER

- Can a detachment ceiling plus a nucleotide-free pool reach the measured post-restretch rate?
- What does it cost elsewhere in the cost function?
- Do the physiological justifications hold against the repo's own α -MHC bounds?
- Is a 9-parameter re-identification worth running?

It reaches the data — and then the k_{tr} overstretch peak falsifies it

It is the only mechanism tested that reaches the measured post-restretch rate...

variant	post-restretch	peak1	E_FV	E_slack
DATA	49.2	1.154		
baseline (defects 1+2 fixed)	92.7	1.266	0.020	180
ceiling 800	100.0	1.460	0.156	964
ceiling 300	184.5	1.620	0.315	3323
ceiling 300 + rip 10000	60.1	1.379	0.310	2079
ceiling 150 + rip 10000	44.8	1.425	0.461	5057

- The ceiling ALONE makes it worse — nothing detaches, peak1 climbs to 1.620 and post-restretch goes to 184.5.
- Tearing is the necessary release valve: 184.5 → 68.9 → 60.1 → 44.8.
- Tearing WITHOUT the ceiling does nothing (92.7 → 86–92) — R2 has already annihilated those bridges. The two only work as a pair.

...but the k_{tr} overstretch peak was measuring R2(s) all along

	overstretch peak (F_iso)	hold at 1.05 ML
DATA 03/27 / 04/03 / 04/10	0.72 / 0.80 / 0.83	0.37 / 0.35 / 0.43
model, NO ceiling	0.746 ✓	0.618
model, ceiling 300	2.249	1.670
model, ceiling 300 + rip 10000	1.183	0.655
model, ceiling 150 + rip 10000	1.586	0.723

- The k_{tr} overstretch drives the model 2.2× further in strain and 4× higher in rate (2989 s⁻¹) than the slack restretch that identified R2(s).
- That force is in the k_{tr} trace and already inside E_ktr as trace MSE — nobody was reading it as a constraint.
- The uncapped model is exact. Every capped variant is 1.4–3× too high.



R2(s) is pinned at BOTH ends of the strain range by data already in the cost — 0.022 μm and 0.050 μm — and the uncapped function matches both. The muscle really does shed force that fast at 25 nm strain.

Three of the four physiological justifications do not survive



“ATP-binding ceiling” is the wrong name — by ~30×

At saturating [MgATP] the ATP-binding sub-step is $>10,000 \text{ s}^{-1}$ apparent for α (Lowey 2013; Deacon 2012). The fitted $k_{2\text{max}}$ is $150\text{--}300 \text{ s}^{-1}$. The $\text{MgATP}/(\text{K}_{\text{T2c}} + \text{MgATP})$ factor is inert: 0.982 at 8 mM vs 0.930 at 2 mM — a 5 % swing where the data wants 2.5×



A ceiling IS defensible — but as ADP release, and only in β territory

$R2(s)$ reaching 4600 s^{-1} is 3–70× above every α -MHC anchor — that much is a real defect. But the settings that WORK ($k_{2\text{max}}$ 150–300) sit below the α isolated-S1 range entirely. The ceilings that stay inside α (800, 2000) make post-restretch worse. It only delivers when pushed outside species-appropriate kinetics.



$k_{\text{DOT}} = 40 \text{ s}^{-1}$ cannot be ATP binding — by ~250×

A 25 ms refractory pool is not an ATP-binding phenomenon at physiological [ATP]. Re-hydrolysis (kah, mouse α 150 s^{-1}) is the right order — but the model already has that step.



D0 works for a bookkeeping reason, not a chemical one

PT (0.11) and PD (0.55) are large pre-filled buffers. D0 starts EMPTY, so it acts as a sink with no reservoir — that, not nucleotide chemistry, is why it slows recovery. Also: $\text{max D0} = 0.59\text{--}0.74$ means 59–74 % of heads torn simultaneously, five cycles in a row, each needing an ATP to re-prime. Lengthening contraction is classically cheap (Abbott & Aubert) — the Fenn effect runs the other way.

WHAT SURVIVES

The structural diagnosis. $R2(s)$ conflates (a) chemical cycle completion — ADP release $\rightarrow$ ATP binding $\rightarrow$ detachment, bounded, ends in M·ATP — with (b) mechanical rupture — unbounded in strain, ATP-INdependent, ends nucleotide-BOUND. The current code gives the rupture branch the chemical branch's destination. Splitting them is right. Not sound: every place [ATP] was invoked.

Verdict: do not run the refit — and what the exercise bought

VERDICT — DO NOT RUN THE CEILING REFIT

The k_{tr} overstretch peak settles it. A ceiling that fixes the post-restretch recovery breaks the overstretch peak by 1.4–3×. Add the 30–250× mismatches in the ATP justifications, and the mechanism is not worth a 9-parameter, ~6 h-per-run re-identification against an 11-DOF identifiability ceiling.

RunOptimApoPool.m is kept as a FALSIFICATION HARNESS, not a fitting plan: it is seeded in bounds and it scores the physiology bounds, so if anyone re-opens this the run will show plainly what it costs.

What the exercise bought, which is not nothing

1

A free, unused constraint

The k_{tr} overstretch peak is a direct measurement of the large-strain detachment. It was sitting inside E_{ktr} as trace MSE, invisible. Promote it to an explicit feature.

2

A much narrower search space

The post-restretch defect cannot live on the $R2(s)$ strain axis — that axis is now measured at both ends.

3

A candidate that costs nothing to test

RegAvailShorteningOnly = 1 means lengthening does not modulate registration availability — and the post-restretch window is precisely a post-lengthening window.

But note — the honest reading of the alternative

A one-sided (lengthening-suppresses) A_{reg} has no mechanism behind it: the site-sweeping argument that motivates the availability boost is direction-blind. And the model already penalises attachment during lengthening — a head attaching at $s=0$ is advected straight into the high-strain region where $R2$ is $700\text{--}3000\text{ s}^{-1}$, so a k_a penalty double-counts. The existing RegAvailShorteningOnly = 1 is itself a fitting device, and should be labelled as one when interpreting fits.

The one argument that survives is not about direction at all: compliant realignment (Daniel 1998; Tanner, Daniel & Regnier 2007) predicts a deficit driven by how much bound mass was disrupted WHILE THE FILAMENT STAYED LOADED — naturally larger after a restretch ($p2\ 0.19 \rightarrow 0.05$ at $0.80\ F_{iso}$) than after a slack ($p2$ collapses, but so does force, to zero). That is the version worth building, if any.



If a mechanism can only be fitted by violating its own bounds — $k2rip$ 6–20× over, 74 % of heads torn — that is the falsification, not a fitting detail.

Where this leaves the project

SETTLED

- ATP effect: force $\times 1.36$, $k_{tr} \times 0.55$. Fit as ratios with a free per-prep scale.
- Rundown: $\phi = 0.45$ on effective activated time; correct force, never k_{tr} .
- Rundown is series-elastic creep — prefer a length offset to a force scale in the model.
- One redevelopment rate, $\sim 41\text{--}47\text{ s}^{-1}$, shared by all three rises.
- Feature reliability is ranked; `ovrsht_*` and `ktr2_*` are retired.
- Baker: keep the k_{tr} , drop the amplitudes.

OPEN

- Mechanistic identification of $\times 1.36 / \times 0.55$ from cross-bridge kinetics — a single k_2 cannot do it (LowATP_k2Frontier).
- Defect 3: post-restretch $\sim 2\times$ too fast, and the R2(s) strain axis is now closed off as a home for it.
- Does $\phi \approx 0.45$ transfer between preparations? Calibrated on one bracket.
- 04/03's t_0 anomaly — lower V_{max} in that prep? Check against that day's FV.
- A per-day passive reference, before passive and active are jointly identified.

CLOSED OFF

- A cap on strain-dependent detachment — falsified by the k_{tr} overstretch peak, on data already in the cost.
- A lengthening penalty on A_{reg} — no mechanism behind the one-sidedness, and it double-counts.
- Slower SRX as a fix for defect 3 — falsified at fixed other parameters (E_{slack} rises 5–9 $\times$).
- Correcting k_{tr} for rundown.
- Fitting absolute kPa to a single preparation.



The model's flexibility is bounded, not infinite — SensitivityAnalysis puts an 11-DOF identifiability ceiling on the Slack/FV/Ktr feature set. Adding parameters does not add constraint.

WHAT I WOULD DO NEXT

Three cheap moves, in order

MODEL

Set UseMaxwellTensionOnly = 1 in ThursdayNightFever.m

Free. k_t , 192 $\rightarrow$ 85.5 s⁻¹, E_ktr 0.98 $\rightarrow$ 0.225, E_slack 183.4 $\rightarrow$ 173.4. FV and stairs unchanged. Strictly an improvement — do it before anything else.

MODEL

Add tau_reg to params0.mods and refit with the k_t trace weighted

Expect it to land in the 1–5 ms range. Do not pin A_reg (E_FV 0.016 $\rightarrow$ 0.168) and do not touch A0 (E_slack $\rightarrow$ 1111). Promote the k_t overstretch peak to an explicit feature in the same round.

COST

Re-baseline the cost function on the survey's six recommendations

Ratios not levels · normalised shape + per-prep scale · exclude Baker amplitudes · retire ovrsh_t* and ktr2_* · detrend before extracting · per-day passive reference.

BENCH

Record a hi-res end-of-session repeat on a second preparation

The single most valuable next measurement: it tests whether $\varphi \approx 0.45$ is universal, and unlocks a direct stiffness measurement (restretchSlopeStart / v_restretch) that separates “fewer heads” from “series creep” in one number.

Analyses/README.md is the index and the cross-analysis synthesis · each folder has its own conclusions.md and results/